

Project Initialization and Planning Phase

Date	05 July 20206
Team ID	**
Project Title	Rising Waters: A Machine Learning Approach for Flood Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution uses Machine Learning to predict flood risks based on historical weather and environmental data. It aims to provide early warnings through a Flask-based web application, helping reduce the impact of floods by enabling timely preventive actions.

Project Overview	
Objective	Develop an AI-based flood prediction system that analyzes weather and river data to provide accurate flood predictions and early alerts.
Scope	The project predicts flood risks using Machine Learning models and displays results through a web application. It supports disaster preparedness and can be extended with real-time APIs and IoT devices.
Problem Statement	
Description	Existing flood warning systems often provide delayed or inaccurate alerts, making it difficult for people and authorities to respond effectively.
Impact	An accurate prediction system helps reduce loss of life, property damage, and improves disaster preparedness.
Proposed Solution	
Approach	Train Machine Learning models on historical flood and weather data and deploy the best-performing model using Flask for real-time flood prediction.
Key Features	• AI-based flood prediction • Multiple ML model comparison (XGBoost, Random Forest, KNN, Decision Tree) • Flask web application • Early warning alerts • User-friendly interface

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Model Training	T4 GPU / Intel i5+
Memory	RAM	8 GB
Storage	Data & Models	50 GB SSD
Software		
Frameworks	Web Development	Flask
Libraries	ML & Data Processing	Scikit-learn, XGBoost, Pandas, NumPy
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Dataset	Historical Flood & Weather Data	Kaggle Dataset (CSV)